Huawei Access Network SOP for MG-ACC-46 MG Road Core Optical Transport Failure

Node: MG-ACC-46 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_8365

Description

The MG-ACC-46 access node failure is attributed to a Huawei optical transport failure, specifically SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. This is due to the proprietary Huawei OTN (Optical Transport Network) protocol stack and the underlying hardware architecture, which relies on the SFP/XFP modules to transmit and receive optical signals. The OTN protocol stack is responsible for framing, multiplexing, and transmitting optical signals over the optical fiber infrastructure. In the event of SFP/XFP degradation, fiber cuts, or Rx/Tx power loss, the OTN protocol stack is unable to maintain the optical signal integrity, resulting in a failure of the MG-ACC-46 access node. Furthermore, the Huawei access node's reliance on the OTN protocol stack and SFP/XFP modules makes it susceptible to optical transport failures, which can have a significant impact on network availability and service quality.

Pre-requisite Checks

- Verify that the remote hands have been dispatched to the MG Road Core site.

- Check the status of the redundant link to ensure it is operational and can take over in the event of a failure.

- Confirm that the alarm code ALARM_8365 is present on the MG-ACC-46 access node.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical transceiver, Expected Output: Transceiver status: OK, Description: This command displays the status of the SFP/XFP module, including any error messages or warnings.

Command: display optical interface, Expected Output: Interface status: DOWN, Description: This command displays the status of the optical interface, including any error messages or warnings.

Command: display fiber fault, Expected Output: Fault status: NO FAULT, Description: This command displays any fiber faults, including any cuts or breaks in the optical fiber.
```

Resolution Steps

Step 1: Step: Replace SFP/XFP module, Command: replace optical transceiver, Description: Replace the SFP/XFP module with a new one to resolve the optical transport failure.

Step 2: Step: Verify fiber integrity, Command: display fiber fault, Description: Verify that the fiber is intact and there are no cuts or breaks.

Step 3: Step: Adjust Rx/Tx power, Command: adjust optical power, Description: Adjust the receive and transmit power levels to ensure optimal signal integrity.

Post-Check Verification

```
Command: display optical transceiver, Expected Output: Transceiver status: OK, Description: Verify that the SFP/XFP module is functioning correctly.

Command: display optical interface, Expected Output: Interface status: UP, Description: Verify that the optical interface is operational.

Command: display fiber fault, Expected Output: Fault status: NO FAULT, Description: Verify that there are no fiber faults.
```

Roll-back Procedures

- Step: Restore previous SFP/XFP module, Command: undo replace optical transceiver, Description: Rollback to the previous SFP/XFP module if the replacement fails.

- Step: Restore previous fiber settings, Command: undo adjust optical power, Description: Rollback to the previous fiber settings if the power adjustment fails.